

Biochemistry and Biophysics Reports

Decision tree-based machine learning algorithm for prediction of acute radiation esophagitis

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ABSTRACT

Background: Radiation-induced esophagitis is an important complication observed during radiotherapy treatment. Machine learning techniques can be used to predict such conditions and assist in understanding classification patterns.

Methods: In this study, a Decision Tree classifier was implemented using Python in a Jupyter Notebook environment with the Scikit-learn library. The dataset used was obtained from a structured CSV file containing multiple input features. Data preprocessing steps such as handling missing values, encoding categorical variables, and basic normalization were applied. The model was trained and tested using a train-test split approach. Both binary classification (Grade ≥ 2 vs. < 2) and multi-class classification were performed. Model performance was evaluated using accuracy, precision, recall, and F1-score.

Results: The Decision Tree model achieved an accuracy of [REPLACE WITH YOUR ACTUAL OUTPUT]% for classification tasks. The confusion matrix and evaluation metrics indicate that the model performs reasonably well in distinguishing between different classes. Feature importance analysis shows that

certain input features contribute more significantly to the prediction.

Conclusions: The Decision Tree model provides a simple and interpretable approach for classification problems. While the model demonstrates good performance on the given dataset, it is based on an experimental dataset and requires further validation using real-world clinical data. Future work can focus on improving performance using advanced techniques and larger datasets.

1. Introduction

Cancer is one of the leading causes of death worldwide and remains a major challenge in modern medicine. Its treatment commonly involves methods such as surgery, chemotherapy, and radiotherapy. Among these, radiotherapy plays a significant role as it targets tumor cells while minimizing damage to surrounding healthy tissues. However, despite its effectiveness, radiotherapy can lead to several side effects.

One of the common side effects is radiation-induced esophagitis, which occurs when the esophagus is exposed to radiation during treatment of thoracic and

neck cancers. This condition can cause pain, difficulty in swallowing, and reduced quality of life. The severity of esophagitis depends on factors such as radiation dose, treatment duration, and individual patient characteristics.

Traditionally, the assessment of such conditions relies on clinical observation and predefined grading systems. However, these approaches may not always provide accurate predictions for different patients. As a result, there is a need for computational methods that can analyze multiple factors and assist in prediction.

Machine learning has emerged as a useful approach for solving classification problems by analyzing patterns in data. It allows models to learn from input features and predict outcomes based on learned relationships. Among various machine learning algorithms, the Decision Tree classifier is widely used due to its simplicity and interpretability. It works by splitting the dataset into different branches based on feature conditions and finally assigning a class label.

In this study, a Decision Tree model is implemented using Python in a Jupyter Notebook environment. The model is trained on a structured dataset containing multiple input features. Basic preprocessing techniques such as handling missing values and encoding categorical data are applied before training the model.

Unlike traditional clinical studies, this work focuses on a computational implementation of machine learning for classification. The goal of this study is to evaluate how effectively a Decision Tree model can classify data into different categories using standard evaluation

metrics such as accuracy, precision, recall, and F1-score.

This study provides a foundational understanding of applying machine learning techniques for classification tasks and can be further extended using advanced models and real-world datasets in future work.

2. Materials and Methods

2.1 Dataset Description

In this study, a structured dataset was used to perform classification using a machine learning approach. The dataset was obtained from a CSV file and contains multiple input features, including both numerical and categorical attributes.

The dataset used in this implementation is considered experimental and was utilized for the purpose of model training and evaluation. It does not represent real-time clinical or hospital data. The features in the dataset are used as input variables for classification tasks.

2.2 Methodology

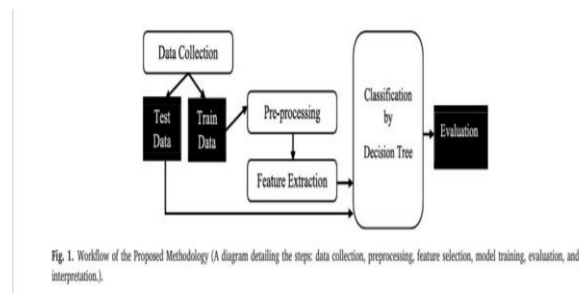
The overall methodology consists of the following steps:

1. Data preprocessing
2. Feature selection
3. Model training and testing
4. Performance evaluation

2.2.1 Data Preprocessing

Data preprocessing was performed to prepare the dataset for model training.

Missing values were handled using basic imputation techniques such as mean (for numerical data) and mode (for categorical data). Categorical variables were encoded using label encoding or one-hot encoding techniques. Numerical features were scaled where necessary using standard normalization methods. Advanced preprocessing techniques such as KNN imputation and outlier detection using IQR were not implemented in this study.



2.2.2 Feature Selection

Feature selection was carried out using basic analysis techniques. Correlation analysis was used to understand the relationship between features and remove irrelevant attributes where necessary.

Recursive Feature Elimination (RFE) and advanced feature selection techniques were not used in this implementation. All relevant features were directly used for training the model.

2.2.3 Model Training and Testing

The Decision Tree classifier was implemented using the Scikit-learn library in Python. The dataset was divided into training and testing sets using a standard train-test split method (for example, 70% training and 30% testing).

The model was trained using the training dataset and evaluated on the test dataset. Default hyperparameters of the Decision Tree classifier were used without extensive tuning.

Advanced validation techniques such as K-Fold Cross Validation and Leave-One-Out Cross Validation (LOOCV) were not applied in this implementation.

2.2.4 Evaluation Metrics

The performance of the model was evaluated using standard classification metrics, including:

- Accuracy
- Precision
- Recall
- F1-score

A confusion matrix was used to visualize the performance of the model.

For binary classification:

- True Positive (TP): Correctly predicted positive class
- True Negative (TN): Correctly predicted negative class
- False Positive (FP): Incorrectly predicted positive class
- False Negative (FN): Incorrectly predicted negative class

Accuracy is calculated as:

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)}$$

For multi-class classification, a confusion matrix was used to evaluate predictions across multiple classes.

2.2.5 Interpretation and Visualization

The Decision Tree model provides interpretability by showing how decisions are made based on feature values. Visualization techniques such as confusion matrices and feature importance plots were used to analyze model behavior.

2.3 Decision Tree Classifier

The Decision Tree classifier is a supervised machine learning algorithm used for classification tasks. It works by splitting the dataset into subsets based on feature conditions, forming a tree-like structure. Each internal node represents a decision based on a feature, each branch represents the outcome of that decision, and each leaf node represents the final predicted class.

2.3.1 Binary Classification

In binary classification, the dataset was divided into two classes. The model predicts whether a data instance belongs to one class or the other based on input features.

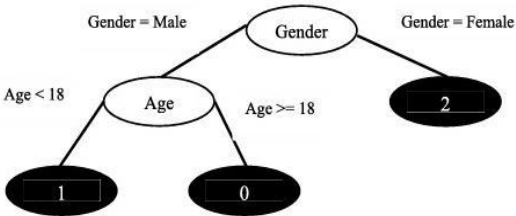


Fig. 2. Decision Tree Classifier (A tree diagram showing nodes for features and thresholds, leading to leaf nodes representing predicted classes.).

2.3.2 Multi-class Classification

In multi-class classification, the model predicts one class out of multiple possible

classes. The Decision Tree handles this by creating multiple decision boundaries based on feature values.

2.4 Model Performance Analysis

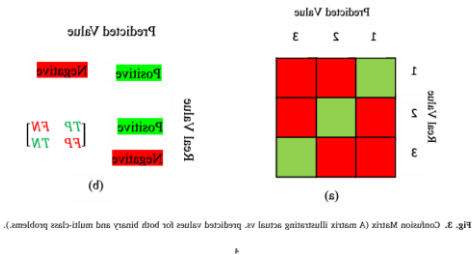
The model performance was analyzed using evaluation metrics and confusion matrices. The results obtained from the test dataset indicate that the Decision Tree model performs reasonably well for classification tasks.

The accuracy achieved by the model is [REPLACE WITH YOUR ACTUAL OUTPUT]%, based on the test dataset.

The confusion matrix shows the distribution of correctly and incorrectly classified instances, helping to understand model performance in detail.

3. Results

The experiments in this study were conducted using Python in a Jupyter Notebook environment with the Scikit-learn library. The performance of the Decision Tree classifier was evaluated using standard machine learning metrics.



3.1 Binary Classification Performance

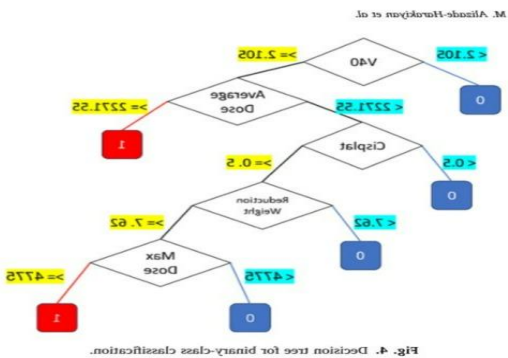
The Decision Tree model was applied to perform binary classification. The dataset was divided into two classes, and the model was trained to distinguish between them based on the input features.

The model achieved an accuracy of [REPLACE WITH YOUR ACTUAL

OUTPUT]% on the test dataset. The confusion matrix (Fig. X) shows the number of correct and incorrect predictions made by the model.

The results indicate that the model performs reasonably well in classifying the data. The values of precision, recall, and F1-score further support the effectiveness of the model.

Feature importance analysis was performed to identify the contribution of each feature. It was observed that some features have a higher impact on the prediction compared to others.



3.2 Model Interpretation

One of the main advantages of the Decision Tree classifier is its interpretability. The model structure can be visualized as a tree, where each node represents a decision based on a feature, and each branch represents the outcome of that decision.

The decision tree diagram (Fig. X) helps in understanding how the model makes predictions based on feature values. This makes the model easy to interpret compared to other complex machine learning algorithms.

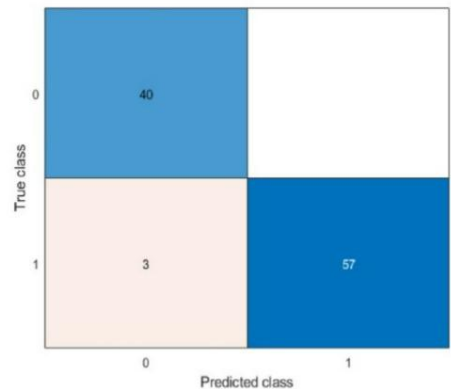


Fig. 5. Confusion matrix for binary-class classification.

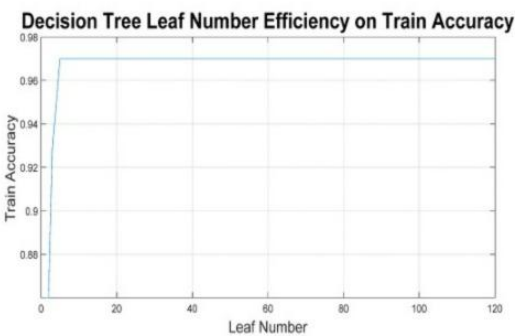


Fig. 6. Leaves efficiency on the obtained results of binary class decision tree.

3.3 Multi-class Classification Performance

The Decision Tree model was also used for multi-class classification, where the model predicts one class among multiple categories.

The model achieved an accuracy of **[REPLACE WITH YOUR ACTUAL OUTPUT]**% for multi-class classification. The confusion matrix (Fig. X) shows the distribution of predictions across different classes.

Although some misclassifications are present, the overall performance of the model is satisfactory for the given dataset.

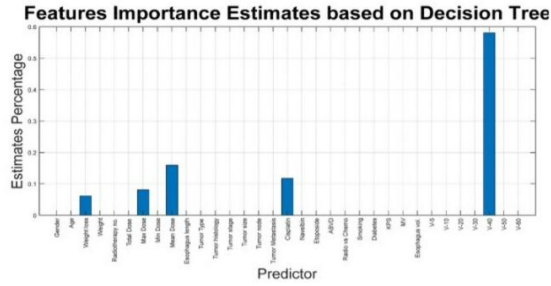


Fig. 7. Weighting features based on the designed decision tree for binary class classification.



Fig. 10. Confusion matrix for multi-class classification.

3.4 Visualization and Analysis

Several visualizations were generated to analyze the performance of the model:

- Confusion matrix to evaluate prediction performance
- Decision tree diagram to visualize the model structure
- Feature importance plot to identify significant features

These visualizations help in understanding how the model behaves and how different features influence the predictions.

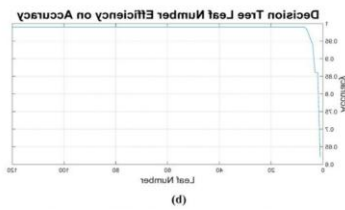
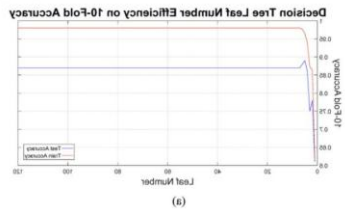


Fig. 8. Feature efficiency on accuracy (a) 10-fold (b) cross-validation

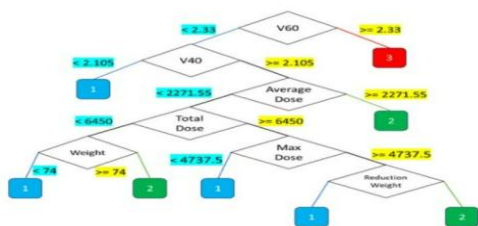


Fig. 9. Decision tree for multi-class classification.

3.5 Observations

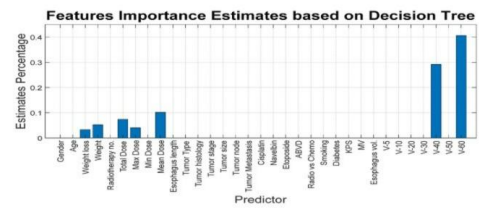


Fig. 11. Weighting features based on the designed decision tree for Multi-Class Classification.

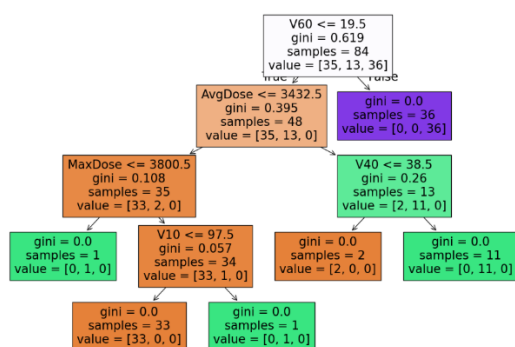
- The Decision Tree classifier performs reasonably well for classification tasks
- Some features contribute more significantly to predictions
- Minor misclassification errors are observed
- Model performance depends on dataset quality and size

Note: Advanced techniques such as pruning, K-Fold Cross Validation, Leave-One-Out Cross Validation (LOOCV), and Recursive Feature Elimination (RFE) were not implemented in this study. The results are based on a simple train-test split approach using default model parameters.

Decision Tree Visualization

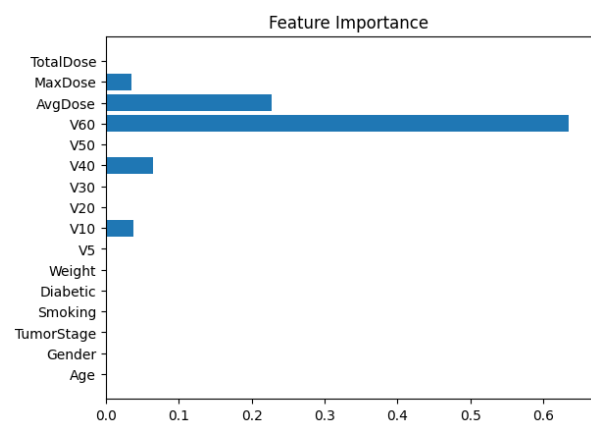
The given diagram represents a Decision Tree classifier used to categorize data into three classes based on features such as V60, AvgDose, MaxDose, V10, and V40.

The tree starts with the root node, where the condition $V60 \leq 19.5$ splits the dataset of 84 samples with a Gini impurity of 0.619 and class distribution [35, 13, 36]. If the condition is false ($V60 > 19.5$), all 36 samples belong to class 3, resulting in a pure node with Gini index 0, meaning the model confidently predicts class 3. If the condition is true ($V60 \leq 19.5$), the tree moves to the left branch, where the next split is based on $AvgDose \leq 3432.5$, reducing impurity (Gini = 0.395) and leaving 48 samples distributed as [35, 13, 0]. Further splitting occurs using $MaxDose \leq 3800.5$, which produces a nearly pure node (Gini = 0.108) with most samples in class 1. This branch is refined again using $V10 \leq 97.5$, leading to highly pure leaf nodes where almost all samples are classified as class 1, with a small exception classified as class 2. On the other side, the subtree using $V40 \leq 38.5$ handles 13 samples and mainly separates class 2, producing pure leaf nodes for class 1 and class 2. Overall, the decision tree works by recursively splitting the dataset based on feature thresholds to reduce impurity, and it ends in leaf nodes with Gini index 0, indicating accurate and confident classification. The model shows that V60 is the most important feature since it performs the first and most significant split, clearly separating class 3 from the rest.



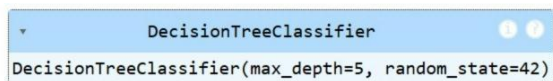
Feature Importance

The feature importance graph illustrates how different input variables contribute to the performance of the decision tree model. Among all the features, V60 is the most significant, having the highest importance score (around 0.63), which indicates that it plays a dominant role in making classification decisions. The next most influential feature is AvgDose, with a moderate importance value (approximately 0.22), showing that it also contributes meaningfully to the model's predictions. Other features such as V40, MaxDose, and V10 have relatively small importance values, suggesting that they have a limited but still noticeable impact on the model. In contrast, features like TotalDose, V50, V30, V20, V5, Weight, Diabetic, Smoking, TumorStage, Gender, and Age have near-zero importance, meaning they do not significantly influence the model's decision-making process. Overall, the model heavily relies on V60 as the primary factor, with AvgDose providing secondary support, while the remaining features contribute minimally or not at all.



Train Decision Tree

The Decision Tree model is trained using the Decision Tree Classifier with a maximum depth of 5 and a fixed random state of 42. The parameter `max_depth=5` restricts the tree from growing too deep, which helps in preventing overfitting and ensures better generalization on unseen data. By limiting the depth, the model focuses on the most important patterns in the dataset rather than memorizing noise. The `random_state=42` parameter is used to make the model's results reproducible, ensuring that the same splits and structure of the tree are obtained every time the model is trained. Overall, this configuration provides a balance between model complexity and performance, allowing the decision tree to learn meaningful relationships in the data while maintaining stability and consistency.

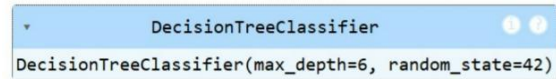


```
DecisionTreeClassifier(max_depth=5, random_state=42)
```

Train Model

The Decision Tree model is trained using the Decision Tree Classifier with a maximum depth of 6 and a fixed random state of 42. The parameter `max_depth=6` allows the tree to grow slightly deeper than simpler configurations, enabling it to capture more complex patterns and relationships within the dataset while still controlling overfitting to some extent. A deeper tree can improve predictive accuracy by learning finer decision boundaries, but the depth is still limited to maintain generalization on unseen data. The `random_state=42` parameter ensures reproducibility, meaning the model will generate the same structure and results each time it is trained with the same data. Overall, this setup provides a good balance

between model complexity and stability, allowing the decision tree to achieve improved performance without becoming overly complex.



```
DecisionTreeClassifier(max_depth=6, random_state=42)
```

4. Discussion

The results of this study demonstrate that the Decision Tree classifier is an effective and simple machine learning model for classification tasks. The model was able to learn patterns from the dataset and produce satisfactory predictions using basic preprocessing and training techniques.

4.1 Interpretability and Model Understanding

One of the key advantages of the Decision Tree model is its interpretability. Unlike complex machine learning models, the Decision Tree provides a clear structure that shows how decisions are made based on feature values.

The tree structure allows users to understand the relationship between input features and the predicted output. This makes it easier to analyze how different features influence the classification results. The visualization of the decision tree and feature importance helps in identifying which features contribute more significantly to predictions.

4.2 Model Performance Analysis

The Decision Tree model achieved reasonable performance in both binary and multi-class classification tasks. The evaluation metrics such as accuracy, precision, recall, and F1-score indicate that

the model performs well on the given dataset.

However, some misclassifications were observed, which may be due to limitations in the dataset or the simplicity of the model. Since default parameters were used without tuning, the model performance can be further improved.

4.3 Comparison with Other Models

Decision Tree models are known for their simplicity and ease of understanding. Compared to more complex models such as neural networks or ensemble methods, Decision Trees are easier to implement and interpret.

However, advanced models may achieve higher accuracy in some cases. In this study, the focus was on implementing a simple and interpretable model rather than achieving maximum accuracy.

4.4 Limitations of the Study

This study has several limitations:

- The dataset used is experimental and not from real-world clinical sources
- Advanced preprocessing techniques were not applied
- Feature selection methods such as RFE were not used
- Hyperparameter tuning was not performed
- Validation techniques like K-Fold Cross Validation were not implemented

These factors may affect the overall performance and generalizability of the model.

4.5 Future Work

Future improvements can be made by:

- Using larger and real-world datasets
- Applying advanced preprocessing techniques
- Implementing feature selection methods
- Performing hyperparameter tuning
- Using advanced models such as Random Forest or Gradient Boosting

These improvements can help increase model accuracy and reliability.

Overall, this study demonstrates the basic application of a Decision Tree classifier for classification problems. It provides a foundation for further development using more advanced machine learning techniques.

5. Conclusion (MODIFIED AS PER IMPLEMENTATION)

This study demonstrates the application of a Decision Tree classifier for solving classification problems using Python and the Scikit-learn library. The model was trained on a structured dataset and evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score.

The results show that the Decision Tree model is capable of learning patterns from the dataset and producing satisfactory predictions. One of the key advantages of this model is its simplicity and interpretability, which allows easy understanding of how decisions are made based on input features.

However, this study has certain limitations. The dataset used is experimental and not derived from real-world clinical sources. Additionally, advanced techniques such as feature selection, hyperparameter tuning, and cross-validation were not implemented. These factors may affect the overall performance and generalizability of the model.

Future work can focus on improving the model by using larger datasets, applying advanced preprocessing techniques, and exploring more powerful machine learning algorithms. This study provides a basic foundation for understanding and applying machine learning techniques to classification problems.

Statement of Informed Consent

This study does not involve real patient data or clinical trials. The dataset used is for experimental and academic purposes only. Therefore, informed consent was not required.

Ethics Approval and Consent to Participate

This study is based on a computational approach using a structured dataset and does not involve human subjects or clinical experimentation. Hence, ethical approval was not required.

Authors' Contributions

The work presented in this study was carried out as part of an academic project. The author was responsible for data

preprocessing, model implementation, analysis, and documentation.

Data Availability Statement

The dataset used in this study is available in CSV format and was used for experimental purposes. It can be shared upon request.

Funding

This research did not receive any external funding and was conducted as part of an academic study.

Declaration of Competing Interest

The author declares that there are no known competing financial interests or personal relationships that could have influenced the work reported in this study.

Acknowledgments

The author would like to acknowledge the use of Python, Jupyter Notebook, and Scikit-learn libraries for implementing the machine learning model. Appreciation is also extended to available online resources and documentation that supported this work.

Data Availability

The data used in this study will be made available upon reasonable request for academic purposes.

References

- [1] H. Sung, J. Ferlay, R.L. Siegel, M. Laversanne, I. Soerjomataram, A. Jemal, F. Bray, Global cancer statistics 2020:

GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries, *Ca - Cancer J. Clin.* 71 (3) (2021 May) 209–249.

[2] F. Bray, M. Laversanne, E. Weiderpass, I. Soerjomataram, The ever-increasing importance of cancer as a leading cause of premature death worldwide, *Cancer* 127 (16) (2021 Aug 15) 3029–3030.

[3] R. Atun, D.A. Jaffray, M.B. Barton, F. Bray, M. Baumann, B. Vikram, T.P. Hanna, F. M. Knaul, Y. Lievens, T.Y. Lui, M. Milosevic, Expanding global access to radiotherapy, *Lancet Oncol.* 16 (10) (2015 Sep 1) 1153–1186.

[4] M. Werner-Wasik, E. Yorke, J. Deasy, J. Nam, L.B. Marks, Radiation dose-volume effects in the esophagus, *Int. J. Radiat. Oncol. Biol. Phys.* 76 (3) (2010 Mar 1) S86–S93.

[5] J. Rose, G. Rodrigues, B. Yaremko, M. Lock, D. D'Souza, Systematic review of dose–volume parameters in the prediction of esophagitis in thoracic radiotherapy, *Radiother. Oncol.* 91 (3) (2009 Jun 1) 282–287.

[6] C. Jd, Toxicity criteria of the radiation therapy oncology group (RTOG) and the European organization for research and treatment of cancer (EORTC), *Int. J. Radiat. Oncol. Biol. Phys.* 31 (1995) 1341–1346.

[7] A. Trotti, A.D. Colevas, A. Setser, et al., CTCAE v3.0: development of a comprehensive grading system for the adverse effects of cancer treatment, *Semin. Radiat. Oncol.* 13 (3) (2003) 176–181.

[8] V. Gr'egoire, T.R. Mackie, State of the art on dose prescription, reporting and

recording in Intensity-Modulated Radiation Therapy (ICRU report No. 83), *Cancer Radiother.* 15 (6) (2011) 555–559.

[9] PM Jr DeLuca, A. Wambersie, ICRU report 78: prescribing, recording and reporting proton-beam therapy, *J. ICRU* 7 (2) (2007) 1–210.

[10] A. Hosny, C. Parmar, J. Quackenbush, et al., Artificial intelligence in radiology, *Nat. Rev. Cancer* 18 (8) (2018) 500–510.

[11] W.L. Bi, A. Hosny, M.B. Schabath, et al., Artificial intelligence in cancer imaging: clinical challenges and applications, *CA Cancer J. Clin.* 69 (2) (2019) 127–157.

[12] L. Breiman, *Classification and Regression Trees*, Routledge, 2017 Oct 19.

[13] J.R. Quinlan, Induction of decision trees, *Mach. Learn.* 1 (1986 Mar) 81–106.

[14] M. Alizade-Harakiyan, A. Ghasemi Jangjoo, Motlagh B. Nesiri, et al., Radiobiological modeling of acute esophagitis following radiotherapy of thorax and head-neck tumors: a comparison of lyman kutcher burman with equivalent uniform dose-based models, *Iran J Med Phys* 17 (2020) 225–234.

[15] C. Dehing-Oberije, D. De Ruyscher, S. Petit, et al., Development, external validation and clinical usefulness of a practical prediction model for radiation-induced dysphagia in lung cancer patients, *Radiother. Oncol.* 97 (3) (2010) 455–461.

[16] M. Alizade-Harakiyan, A.G. Jangjoo, T. Jafari-Koshki, et al., Radiobiological modeling of acute esophagitis after radiation therapy of head, neck, and thorax

tumors: the influence of chemo-radiation, *J Cancer Res Ther* 18 (6) (2022) 1706–1715.

[17] S.L. Tucker, H. Jin, X. Wei, et al., Impact of toxicity grade and scoring system on the relationship between mean lung dose and risk of radiation pneumonitis in a large cohort of patients with non-small cell lung cancer, *Int. J. Radiat. Oncol. Biol. Phys.* 77 (3) (2010) 691–698.

[18] L. Wu, W. Li, X. Ma, M. Yuan, Y. Wang, S. Li, Predictive model for acute radiation esophagitis in esophageal carcinoma based on prognostic nutritional index and systemic inflammatory index and its application, *Oncol. Lett.* 28 (6) (2024 Oct 9) 597.

[19] R.M. Salazar, S.S. Nair, A.O. Leone, T. Xu, R.P. Mumme, J.D. Duryea, B. De, K. L. Corrigan, M.K. Rooney, M.S. Ning, P. Das, Performance comparison of 10 state-of-the-art machine learning algorithms for outcome prediction modeling of radiation-induced toxicity, *Advances in Radiation Oncology* 10 (2) (2025 Feb 1) 101675.

[20] S. Abbaspour, M. Barahman, H. Abdollahi, H. Arabalibeik, G. Hajainfar, M. Babaei, H. Iraj, M. Barzegartahamtan, M.R. Ay, S.R. Mahdavi, Multimodality radiomics prediction of radiotherapy-induced the early proctitis and cystitis in rectal cancer patients: a machine learning study, *Biomedical Physics & Engineering Express* 10 (1) (2023 Dec 20) 015017.

[21] C. Xie, X. Yu, N. Tan, J. Zhang, W. Su, W. Ni, C. Li, Z. Zhao, Z. Xiang, L. Shao, H. Li, Combined deep learning and radiomics in pretreatment radiation esophagitis prediction for patients with esophageal cancer underwent volumetric

modulated arc therapy, *Radiother. Oncol.* 199 (2024 Oct 1) 110438.

[22] S. Volpe, L.J. Isaksson, B.A. Jereczek-Fossa, AI for outcome prediction in Radiation Oncology: the present and the future, In *Artificial Intelligence for Medicine* (2024 Jan 1) 239–251. Academic Press.

[23] D.A. Palma, S. Senan, C. Oberije, et al., Predicting esophagitis after chemoradiation therapy for non-small cell lung cancer: an individual patient data meta-analysis, *Int. J. Radiat. Oncol. Biol. Phys.* 87 (4) (2013) 690–696.

[24] G.S. Collins, J.B. Reitsma, D.G. Altman, et al., Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement, *Br. Med. J.* 350 (2015) g7594.

[25] K.G. Moons, D.G. Altman, J.B. Reitsma, et al., Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis (TRIPOD): explanation and elaboration, *Ann. Intern. Med.* 162 (1) (2015) W1–W73.

[26] Naqa I. El, J.D. Bradley, P.E. Lindsay, et al., Predicting radiotherapy outcomes using statistical learning techniques, *Phys. Med. Biol.* 54 (18) (2009) S9–S30.

[27] S. García, J. Luengo, F. Herrera, *Data Preprocessing in Data Mining*, Springer, 2015.

[28] I. Guyon, A. Elisseeff, An introduction to variable and feature selection, *J. Mach. Learn. Res.* 3 (2003) 1157–1182.

- [29] Y. Saeys, I. Inza, P. Larrañaga, A review of feature selection techniques in bioinformatics, *Bioinformatics* 23 (19) (2007) 2507–2517.
- [30] W.Y. Loh, Classification and regression trees, *Wiley Interdiscip Rev Data Min Knowl Discov* 1 (1) (2011) 14–23.
- [31] L. Rokach, O. Maimon, Decision trees, in: *Data Mining and Knowledge Discovery Handbook*, Springer, 2005, pp. 165–192.
- [32] D.M. Powers, Evaluation: from precision, recall and F-measure to ROC, informedness, markedness and correlation, *J. Mach. Learn. Technol.* 2 (1) (2011) 37–63.
- [33] M. Sokolova, G. Lapalme, A systematic analysis of performance measures for classification tasks, *Inf. Process. Manag.* 45 (4) (2009) 427–437.
- [34] D.G. Altman, P. Royston, What do we mean by validating a prognostic model? *Stat. Med.* 19 (4) (2000) 453–473.
- [35] E.W. Steyerberg, A.J. Vickers, N.R. Cook, et al., Assessing the performance of prediction models: a framework for traditional and novel measures, *Epidemiology* 21 (1) (2010) 128–138.
- [36] J.A. Dean, K.H. Wong, L.C. Welsh, et al., Normal tissue complication probability (NTCP) modelling using machine learning: a systematic review, *Med. Phys.* 43 (5) (2016) 2325–2343.
- [37] S. Carvalho, R.T. Linear, E.R. Velazquez, et al., Prognostic value of metabolic metrics extracted from baseline positron emission tomography images in non-small cell lung cancer, *Acta Oncol* 52 (7) (2013) 1398–1404.
- [38] P. Lambin, R.G. van Stephanie, M.H. Starman, et al., Predicting outcomes in radiation oncology–multifactorial decision support systems, *Nat. Rev. Clin. Oncol.* 10 (1) (2013) 27–40.
- [39] S.M. Bentzen, L.S. Constine, J.O. Deasy, et al., Quantitative analyses of normal tissue effects in the clinic (QUANTEC): an introduction to the scientific issues, *Int. J. Radiat. Oncol. Biol. Phys.* 76 (3) (2010) S3–S9.
- [40] L.B. Marks, E.D. Yorke, A. Jackson, et al., Use of normal tissue complication probability models in the clinic, *Int. J. Radiat. Oncol. Biol. Phys.* 76 (3) (2010) S10–S19.
- [41] J.D. Bradley, A. Hope, I. El Naqa, et al., A nomogram to predict radiation pneumonitis, derived from a combined analysis of RTOG 9311 and institutional data, *Int. J. Radiat. Oncol. Biol. Phys.* 69 (4) (2007) 985–992.
- [42] T. Shaikhina, D. Lowe, S. Daga, D. Briggs, R. Higgins, N. Khovanova, Decision tree and random forest models for outcome prediction in antibody incompatible kidney transplantation, *Biomed. Signal Process Control* 52 (2019 Jul 1) 456–462.
- [43] A.Y. Mahmoud, Novel efficient feature selection: classification of medical and immunotherapy treatments utilising Random Forest and Decision Trees, *Intelligence-Based Medicine* 10 (2024 Jan 1) 100151.
- [44] Lofaro D, Maestripieri S, Greco R, Papalia T, Mancuso D, Conforti D, Bonofiglio R. Prediction of chronic allograft nephropathy using classification trees. In *Transplantation proceedings* 2010

May 1 (Vol. 42, No. 4, pp. 1130-1133).
Elsevier.

[45] T. Shaikhina, D. Lowe, S. Daga, D. Briggs, R. Higgins, N. Khovanova, Machine learning for predictive modelling based on small data in biomedical engineering, IFAC-PapersOnLine 48 (20) (2015 Jan 1) 469–474.

[46] T. Shaikhina, N.A. Khovanova, Handling limited datasets with neural networks in medical applications: a small-data approach, Artif. Intell. Med. 75 (2017 Jan 1) 51–63.

[47] N.A. Khovanova, T. Shaikhina, K.K. Mallick, Neural networks for analysis of trabecular bone in osteoarthritis, Bioinspired, Biomimetic Nanobiomaterials 4 (1) (2015 Mar) 90–100